

Sairam Bokka

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SKILLS

Languages: Python, TypeScript, JavaScript, HTML5, CSS, Bash

Frontend: React, Next.js, Tailwind CSS, Context API, React Hooks

Backend: Node.js, Express.js, FastAPI, RESTful APIs, WebSocket, Socket.IO

Infrastructure & Tools: Docker, Docker Compose, Jenkins, Git, CI/CD, Linux, Playwright

AI & Security: LangChain, CrewAI, PyTorch, Scapy, CompTIA Security+, Splunk, Nmap

WORK EXPERIENCE

Corca Research Inc

Remote

Software Engineering Intern

Dec 2025 – Present

- Expanding Phantom Hints (tab-to-autocomplete) functionality for a SymPy-powered math engine deployed on the frontend via WebAssembly, broadening hint coverage across integrals, derivatives, functions, sets, and other expression types.
- Engineered support for 20+ new mathematical hint categories by extending the SymPy parsing and suggestion pipeline, improving autocomplete accuracy for complex symbolic math workflows.
- Integrated the WebAssembly module on the frontend to ensure seamless rendering and low-latency hint delivery within the browser-based math editor.

RootsID LLC

Baltimore, MD

Software Engineering Intern

Feb 2025 – May 2025

- Refactored JavaScript/TypeScript pilot codebase into 5+ reusable Node.js packages with modular architecture following OOP principles, reducing development time by 30% and eliminating 200+ lines of duplicate code.
- Orchestrated containerized deployment for 4+ microservices across distributed Linux environments, optimizing inter-service communication and reducing system setup latency by 25% through automated Docker Compose workflows.
- Debugged and resolved 15+ critical issues across 3 Node.js microservices using Git version control and systematic testing, reducing system crashes by 30% and improving user verification success rate.
- Developed secure file upload feature using JavaScript and RESTful API design, enabling 100+ users to process and sign documents in-browser with KERI-based credentials.

Tenable Network Security

Baltimore, MD

Security Engineering Intern

June 2024 – August 2024

- Developed Python-based Slack bot with asynchronous RESTful API integration for automated lookups across 8 repositories, reducing response time from 3–5 days to under 24 hours and improving efficiency by 80%.
- Containerized application using Docker with comprehensive unit testing in Python, implementing 50+ tests, achieving 95% code coverage for production reliability.
- Built Jenkins-based CI/CD pipeline integrated with Git for automated builds, testing, and deployment, reducing manual effort by 40% and supporting scalable rollout of internal tools.

Concentrix Catalyst

Hyderabad, IN

Software Engineer

March 2022 – June 2023

- Contributed to the core architecture of 4 full-stack applications, delivering high-impact features across the complete product lifecycle.
- Collaborated with a 5-person cross-functional team to build a dynamic article landing page system in Adobe Experience Manager (AEM).
- Created modular AEM templates that empowered 8+ non-technical authors to launch high-traffic landing pages independently through dynamic JSON data retrieval.
- Optimized user experience by implementing asynchronous content loading via AJAX, ensuring seamless data rendering and faster page load speeds.

PROJECTS

Financial Analyzer | Next.js, TypeScript, Claude AI

- Built a full-stack financial analysis platform leveraging the Claude AI API to process and interpret financial data, delivering AI-generated insights through a responsive Next.js frontend.
- Containerized the application with Docker for deployment, and structured a clean TypeScript codebase with modular components.

Real-Time Scalable Chat Application | React, Socket.IO, Node.js

September 2025

- Developed a high-concurrency messaging platform featuring bidirectional WebSocket communication with <100ms latency
- Implemented JWT-based session management and Google OAuth2 to ensure secure, persistent user authentication

EDUCATION

University of Maryland, Baltimore County

Baltimore, Maryland

Master of Science in Cybersecurity, GPA 4.0/4.0

May 2025

Guru Nanak Institutions Technical Campus

Hyderabad, India

Bachelor of Technology in Electronics and Communications Engineering, GPA 3.6/4.0

August 2022